First Progress Review

Chapter 2: Literature Review

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

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Draft version 1 · 10 November 2026

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2. Literature Review

This chapter reviews the literature on which the framework builds and identifies, for each objective, the point at which existing work stops. It updates the review in the accepted proposal with the Research Paper Bank assembled during the first months of candidature. The chapter first describes how the literature was searched, selected and recorded (Section 2.1), then reviews the clinical and data context (Section 2.2) and each objective in turn: multi-agent orchestration (O1, Section 2.3), uncertainty and deferral (O2, Section 2.4), data-efficient, multi-source and federated models (O3, Section 2.5) and clinical evaluation of human-AI teams (O4, Section 2.6). Section 2.7 synthesises what is established and what remains open.

2.1 Review Method

2.1.1 Search and selection

The review is a structured narrative review organised by objective. Searches were run in PubMed and PubMed Central, arXiv, Google Scholar, IEEE Xplore and the ACM Digital Library, and the proceedings of MICCAI, NeurIPS, ICML, ICLR, CVPR and AAAI were browsed directly. Search strings combined a domain term (brain tumour, glioma, neuro-oncology, medical imaging, computational pathology, clinical decision support) with a method term for each objective: multi-agent, orchestrator, agentic, tool use (O1);

calibration, uncertainty, selective prediction, learning to defer, conformal prediction, abstention, out-of-distribution (O2); self-supervised, transfer learning, missing modality, multiple-instance learning, foundation model, federated learning, differential privacy (O3); and human-AI collaboration, reader study, automation bias, workload, usability, trust, reporting guideline (O4). Reference lists of key papers were followed backwards and forwards.

Inclusion criteria were: (i) relevance to at least one objective; (ii) publication from 2021 onwards, except for seminal methods that define a field (for example temperature scaling, deep ensembles, selective classification and learning to defer); (iii) peer-reviewed venue or a preprint from an established group, with predatory journals excluded; (iv) an open-access full text, so that every paper can be read and re-checked; and (v) enough methodological detail to extract the study design, data and evaluation. As advised at the supervision meeting of 5 November 2026, priority was given to papers that describe their methodology in detail, to studies with external validation, and to studies reporting negative results.

The collection grew in two rounds. By 9 November 2026, 314 papers had been collected and their methodology extracted. A second round on 10 November 2026 added 313 open-access papers and removed 68 that had no open-access full text, were duplicates, or no longer matched the objectives, giving the current bank of **559 papers** (Table 2.1). Of these, 524 (94%) were published in 2021 or later, 299 appeared in journals or conference proceedings and 260 are arXiv or medRxiv preprints; 13 were designated *base papers* whose design the framework adopts or extends directly.

Table 2.1: The Research Paper Bank by objective (as of 10 November 2026).

Objective	Theme	Papers	Base papers
O1	Multi-agent architecture, orchestration, uncertainty hand-off and audit trail	143	4
O2	Calibrated uncertainty and escalation (when to defer to the clinician)	109	3
O3	Data-efficient MRI and pathology models, fusion and federated learning	146	4
O4	Clinical evaluation: performance, escalation appropriateness, workload, usability, trust and safety	161	2
Total		559	13

2.1.2 Data extraction

For every paper a structured record was completed in a review sheet per objective: aim, method, main result, tools, research gap, relation to this study and limitations. For papers

with full methodological reporting (246 so far), a detailed extraction was added covering study design and institutions, datasets and sample sizes, preprocessing, model architecture, training settings, baselines, evaluation metrics, statistical tests and availability of code and data. These records are published in the Research Paper Bank on the progress website, where each paper links to its source. The extraction directly informs the methodology in Chapter 3: baselines, metrics and statistical tests were chosen from what comparable studies used.

2.2 Clinical and Data Context

Brain and nervous-system tumours are uncommon but carry disproportionate mortality (Price et al., 2024). Sri Lanka sees roughly 700 new cases and 587 deaths a year (International Agency for Research on Cancer, 2024; National Cancer Control Programme, 2024). The most consequential recent change for computational work is the 2021 WHO classification, which integrates histology with molecular markers to define tumour type and grade (Louis et al., 2021). A model trained on a pre-2021 label is therefore not predicting the same target as one aligned to the current classification, and single-source systems can only support a diagnosis, never produce one of record. Surgery is constrained by the trade-off between extent of resection and preservation of eloquent function (Sanai & Berger, 2008; Molinaro et al., 2020).

The data on which most published models are built differ from the Sri Lankan setting. BraTS 2021 provided about 2,000 multi-institutional glioma cases, but mostly from Western centres and in pre-processed, co-registered form (Baid et al., 2021). BraTS-Africa was created to test whether such models work on routine-care MRI from Sub-Saharan Africa, acquired on older or low-field scanners with poorer contrast and resolution in patients who present later (Adewole et al., 2026); the winning team reached a lesion-wise Dice of 0.86 only after segmentation-aware augmentation and ensembling of three architectures on the small local dataset (Ankomah et al., 2025). Low- and middle-income settings also face shortages of scanners, specialists and pathology services, delayed presentation and poor incidence data (Yevudza et al., 2024), and AI evaluations there are few and rarely real-world (Ciecierski-Holmes et al., 2022). These observations motivate the data-efficient, locally validated and federated approach of O3.

2.3 Multi-Agent Orchestration in Clinical AI (O1)

2.3.1 From single models to agents

Agentic AI combines a reasoning core with tools, memory and the ability to act over several steps. Foundational work showed that interleaving reasoning with tool calls improves reliability (Yao et al., 2023), and general frameworks such as AutoGen and MetaGPT made multi-agent conversation and role assignment easy to build (Wu et al., 2023; Hong et al., 2023). A recent architecture for healthcare agents describes four components (planning, action, reflection and memory) and proposes evaluation on recommendation accuracy, task execution, adaptation and explainability (Liu et al., 2025).

2.3.2 Medical multi-agent systems

Medical multi-agent systems fall into three groups. The first simulates a multidisciplinary team with language-model agents. MDAgents lets a moderator agent rate case complexity and assign a single agent, a team or an integrated care team, achieving the best accuracy on seven of ten medical benchmarks (Kim et al., 2024); MedAgents and the Multi-Agent Conversation framework showed similar gains from structured discussion, the latter on 302 rare-disease cases (Tang et al., 2024; Chen et al., 2025). The second orchestrates specialised tools. An oncology agent that called vision transformers, MedSAM, OncoKB and guideline retrieval raised decision accuracy on 20 multimodal cases from about 30% to 87% (Ferber et al., 2025); MedRAX and CXRAgent coordinate chest X-ray tools, CXRAgent adding an evidence validator that rates each tool claim as high, medium or low confidence (Fallahpour et al., 2025; Lou et al., 2026). The third is closest to this study: orchestrated clinical workflows with a defined pathway. An orchestrator with seven red-flag agents supported secondary-headache diagnosis in primary care (Wu et al., 2026); an emergency triage system used a weighted confidence score below 0.7 to raise acuity, keeping the clinician as final decision-maker (Sharma et al., 2026); a multi-agent system automated BT-RADS scoring for glioma follow-up MRI with 76.0% agreement with an expert neuroradiologist, against 57.5% for the original clinical reports (Jabal et al., 2026); and TumorBoard combined radiology, neuropathology, molecular and guideline agents with a claim-evidence ledger and a safety governor that deferred 84.2% of unsafe cases under evidence deletion (Liu et al., 2026).

At the most autonomous end, an agent operating in a simulated, FHIR-based electronic health record managed 574 emergency cases with 88.9% diagnostic accuracy and

outperformed board-certified physicians on matched cases (Ferber et al., 2026). Under clinical-scale workloads, orchestrated multi-agent designs sustained accuracy far better than a single agent handling all tasks (65.3% versus 16.6% at 80 concurrent tasks) and were cheaper to run (Klang et al., 2025).

2.3.3 Failure, consensus and accountability

The same literature documents how such systems fail. An analysis of 1,642 execution traces from seven frameworks found failure rates of 41% to 86.7%, caused by system design (44.2%), inter-agent misalignment (32.3%) and weak verification (23.5%) (Cemri et al., 2025). An audit of medical multi-agent systems over 14,400 cases found that unsupported observations entered 16.6% of cases and propagated downstream, that agents simply repeated their initial view in 98.4% of discussions, and that final answers showed authority bias in 28.8% (Zhu et al., 2025). Optimising each component separately does not optimise the system: a best-of-breed system with excellent information-gathering scored lower diagnostic accuracy (67.7%) than an integrated multi-agent design (77.4%) (Bedi et al., 2025). Language models alone also remain unreliable for clinical decisions, misreading laboratory results and failing to follow guidelines (Hager et al., 2024). Governance work provides documentation and audit frameworks—model cards (Mitchell et al., 2019), internal algorithmic auditing (Raji et al., 2020) and the FUTURE-AI consensus (Lekadir et al., 2025)—but these are organisational processes rather than run-time mechanisms.

2.3.4 Gap for O1

Existing medical multi-agent systems route by question difficulty, role or language-model self-assessment; confidence, where present, is qualitative or heuristic (Lou et al., 2026; Sharma et al., 2026). Only a handful of recent preprints attach statistically grounded uncertainty to routing (Li et al., 2026; Hauke et al., 2026), and they address a single routing step on chest radiographs. No published architecture for the brain-tumour pathway makes calibrated uncertainty part of the contract between agents, propagates it across dependent hand-offs, and records every AI output and clinician validation in an auditable trail. This is the contribution addressed by O1 and RQ1.

Table 2.2: Representative multi-agent clinical systems and what they lack relative to this study.

Study	Domain and design	Main result	Missing for this study
Kim et al. (2024)	LLM agents; moderator assigns team size by complexity	Best on 7/10 medical QA benchmarks	No calibrated confidence; no clinician in the loop
Ferber et al.	Oncology; GPT-4 calling	Decision accuracy ~30% to	No deferral rule; tool errors

Study	Domain and design	Main result	Missing for this study
(2025)	vision and retrieval tools	~87% (20 cases)	pass silently
Lou et al. (2026)	Chest X-ray; director with evidence validator	67.0% CheXbench; preferred by radiologists	Confidence is High/Medium/Low, not calibrated
Sharma et al. (2026)	Emergency triage; orchestrator, four agents	80% exact ESI match, 0% undertriage (60 vignettes)	Heuristic weights; simulated cases only
Jabal et al. (2026)	Glioma follow-up MRI; extractor and scorer agents	76.0% vs 57.5% BT-RADS agreement (492 exams)	No uncertainty or escalation
Liu et al. (2026)	Neuro-oncology tumour board; evidence ledger, safety governor	Deferred 84.2% of unsafe cases	Preprint; synthetic benchmark; no imaging models
Li et al. (2026)	Chest X-ray routing with conformal risk control	Best risk-coverage trade-off	Single step; no dependent chain or clinician cost

2.4 Uncertainty, Calibration and Deferral (O2)

2.4.1 Estimating uncertainty

The premise of the framework is that an agent must know when it does not know. Monte Carlo dropout approximates Bayesian inference in deep networks (Gal & Ghahramani, 2016) and deep ensembles provide a simple, strong alternative (Lakshminarayanan et al., 2017); separating aleatoric from epistemic uncertainty clarifies which part more data can reduce (Hüllermeier & Waegeman, 2021). Reviews of uncertainty quantification in medical deep learning group methods into Bayesian, ensemble, test-time augmentation, deterministic single-pass and post-hoc families (Lambert et al., 2024; Loftus et al., 2022). In brain-tumour segmentation, the QU-BraTS challenge ranked uncertainty methods and found ensembles and test-time augmentation common among the best teams (Mehta et al., 2022), segment-level uncertainty tracked Dice and could flag poor segmentations (Mehrtash et al., 2020), and conformal prediction produced voxel-wise uncertainty that correlated strongly with Dice ($r = -0.83$) on internal and external glioma data (Elyassirad et al., 2025). For IDH prediction from MRI, a confidence-informed network trained on 2,481 gliomas from eight institutions reached 96% accuracy with a Brier score of 0.013 (Bangalore Yogananda et al., 2025). In pathology, uncertainty-thresholded predictions raised external AUROC from 0.93 to 0.99 (Dolezal et al., 2022), and an uncertainty-aware ensemble separated glioblastoma from lymphoma and flagged 67 unseen tumour types with AUROC 0.92 where Monte Carlo dropout reached only 0.67 (Zhao et al., 2025).

2.4.2 Calibration and its fragility

Modern networks are over-confident, and temperature scaling corrects most of this on in-distribution data (Guo et al., 2017). The central negative result is that calibration degrades

under dataset shift, exactly where it is most needed; temperature scaling fitted on clean validation data does not hold, and deep ensembles are the most robust option (Ovadia et al., 2019). Medical benchmarks confirm the problem: no confidence-scoring function reliably prevented silent failures in medical image classification, and none consistently beat the plain maximum softmax response (Bungert et al., 2023; Bernhardt et al., 2022; Jaeger et al., 2023). In pathology, ensembles gave the best uncertainty under domain shift and label noise (Mehrtens et al., 2023b). Multi-domain temperature scaling, which uses heterogeneity across sites, improves robustness to shift (Yu et al., 2022), a result directly relevant to a federation of a few heterogeneous hospitals.

2.4.3 Acting on uncertainty: selective prediction, deferral and conformal guarantees

Selective classification gives a model a reject option and guarantees the error on accepted cases: on ImageNet, 2% top-5 error was guaranteed at about 60% coverage (Geifman & El-Yaniv, 2017). Calibrated selective classification further requires the accepted cases to be well calibrated, including under a new hospital (Fisch et al., 2022). Learning to defer goes further by modelling the human expert: the system learns a predictor and a rejector jointly with a consistent surrogate loss (Mozannar & Sontag, 2020; Madras et al., 2018), later made calibrated with one-vs-all classifiers (Verma & Nalisnick, 2022) and extended to multiple experts, cost-sensitivity and workload constraints (Verma et al., 2023; Alves et al., 2024). Triage formulations allocate scarce human effort explicitly (Raghu et al., 2019; Okati et al., 2021), and sequential deferral has been studied in simulated treatment settings (Joshi et al., 2022). In medicine, combining deferral with uncertainty reached the same F1 while deferring far fewer patients (for example 36% versus 69% for pleural effusion) (Liu et al., 2022), and complementarity-driven deferral to clinicians reduced false positives and clinician reads in real screening programmes (Dvijotham et al., 2023).

Conformal prediction provides distribution-free coverage guarantees for any model (Angelopoulos & Bates, 2021), and conformal risk control extends them to arbitrary monotone losses such as the false-negative rate (Angelopoulos et al., 2024a). Medical applications have flagged unreliable predictions in prostate pathology (one error in 794 predictions versus fourteen without conformal prediction) (Olsson et al., 2022), reduced false positives of a head-CT model at a new hospital from 45% to 5% while abstaining on 14% of examinations (Angelopoulos et al., 2024b), and combined conformal selection with cost-aware deferral for sepsis triage under shift (Kwon & Kim, 2026). In brain-tumour MRI classification, withholding 1.8% of predictions cut selective risk by a third

(Alshamlan, 2026). The guarantees, however, are marginal rather than per patient, assume exchangeable data, and can fail for subgroups and under shift (Mehrtens et al., 2023a); federated conformal prediction is only beginning to address distributed sites (Lu et al., 2023).

2.4.4 Gap for O2

Three limitations define the gap. These formulations address one model making one decision, not a chain in which uncertainty must be composed across agents. They assume a fixed, homogeneous cost of error, whereas clinical costs are asymmetric and differ by decision type. And they model the expert as always available, whereas specialist time is the binding constraint (Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020; Alves et al., 2024). Large-language-model agents add a further difficulty: the usual aleatoric/epistemic split does not transfer to interactive agents (Kirchhof et al., 2025), and how uncertainty is *communicated* to a downstream agent changes whether it is used (Hauke et al., 2026). A calibrated escalation policy over a dependent chain, with clinically elicited asymmetric costs and finite capacity, is the contribution of O2 and RQ2.

2.5 Data-Efficient, Multi-Source and Federated Models (O3)

2.5.1 MRI segmentation and classification

U-Net-based architectures remain the backbone of brain-tumour segmentation. nnU-Net configures itself to each dataset and matched or beat specialised methods on 23 public datasets (Isensee et al., 2021); a later re-evaluation found that many claimed improvements over it disappear under rigorous validation (Isensee et al., 2024). Transformer designs such as Swin UNETR and TransBTS (Hatamizadeh et al., 2022; Wang et al., 2021) and scaled convolutional networks such as MedNeXt (Roy et al., 2023) are competitive, and promptable foundation models (MedSAM, MedSAM2) generalise across modalities but need prompts and struggle with small or unclear boundaries (Ma et al., 2024; Ma et al., 2025). On real-world clinical data with missing sequences, nnU-Net remained within or above the range of human inter-rater agreement (Dice 0.86 internal, 0.93 external) (Pemberton et al., 2023). Routine scans often lack sequences; dedicated models learn shared representations that tolerate missing modalities, with the largest gains when few sequences are available (Zhang et al., 2022; Liu et al., 2023).

2.5.2 Learning with few labels

Because annotated local data will be scarce, label efficiency is essential. Self-supervised pre-training of Swin transformers on unlabelled 3D volumes improved downstream segmentation after fine-tuning (Tang et al., 2022), and stepwise transfer learning achieved expert-level paediatric tumour segmentation from limited data, with AI contours judged acceptable more often than expert contours (80.2% versus 65.4%) (Boyd et al., 2024). Even limited target-domain adaptation yields rapid gains in cross-dataset brain-tumour segmentation (Grande et al., 2026). What remains unestablished is whether representations learned this way produce confidence estimates fit to drive an autonomy decision; most evaluations report accuracy only.

2.5.3 Computational pathology and fusion

Attention-based multiple-instance learning lets a single slide-level label supervise attention over thousands of tiles (Ilse et al., 2018), and CLAM made this data-efficient and interpretable (Lu et al., 2021). Pathology foundation models such as UNI now serve as general-purpose encoders with strong few-shot transfer (Chen et al., 2024b), although clinical benchmarks show that their performance varies by task and site (Campanella et al., 2025) and that they do not remove site-specific batch effects (Kömen et al., 2024). In brain tumours, deep learning reached neuropathologist-level integrated classification of adult diffuse gliomas (AUC above 0.90 for all major types) (Wang et al., 2023), and intra-operative molecular classification has been achieved with stimulated Raman histology and ultra-fast sequencing, the latter abstaining rather than making confident errors (Hollon et al., 2023; Vermeulen et al., 2023). Fusion of histology with genomic features outperformed either source and WHO 2016 grading (Chen et al., 2022), and a mixture-of-experts fusion of pathology and radiology foundation models reached AUC 0.94 on an independent test set for WHO 2021 subtyping while still working when only one modality was present (Saueressig et al., 2026). Multimodal radiology-pathology-text benchmarks are beginning to appear (Rivera et al., 2026).

2.5.4 Federated and privacy-preserving learning

Federated averaging trains a shared model by exchanging parameters rather than data (McMahan et al., 2017); FedProx and SCAFFOLD correct for heterogeneous clients (Li et al., 2020; Karimireddy et al., 2020), and FedBN keeps batch normalisation local to absorb scanner differences (Li et al., 2021). Early medical studies showed feasibility without

sharing patient data (Sheller et al., 2020), and the decisive demonstration trained a glioblastoma boundary model across 71 sites on six continents (Pati et al., 2022). The FeTS challenges benchmarked aggregation methods and exposed worst-case sites hidden by good average Dice (Zenk et al., 2025; Linardos et al., 2025). An extensive benchmark found that plain FedAvg already performs very well for brain-tumour segmentation, with other methods giving only small gains (Manthe et al., 2024). An international paediatric platform lost less than 1.5% in classification and 3% in segmentation relative to centralised training (Lee et al., 2024), and federated evaluation platforms such as MedPerf allow benchmarking at sites without moving data (Karargyris et al., 2023). Personalised methods address appearance shift and modality mismatch between hospitals (Jiang et al., 2022; Wang et al., 2022; Dai et al., 2024), and differentially private federated learning has been shown to be viable for medical images (Adnan et al., 2022; Abadi et al., 2016).

2.5.5 Gap for O3

Three gaps remain. Most fusion methods assume complete, high-quality inputs, whereas in practice histology lags imaging, slide quality varies and molecular markers may be unaffordable. Label-efficient and federated methods are validated almost entirely on Western, curated cohorts and large, well-resourced sites; the regime of a handful of small, heterogeneous hospitals is little characterised. And federation is studied as a training-time issue in isolation: how aggregation changes calibration, and therefore the behaviour of a deferral policy validated on an earlier model, is essentially unexamined (Koutsoubis et al., 2024; Koutsoubis et al., 2025). These define O3 and RQ3.

2.6 Clinical Evaluation of Human-AI Teams (O4)

2.6.1 Reporting and validation standards

Reporting guidelines now cover each stage of clinical AI evaluation: DECIDE-AI for early-stage live evaluation of decision support, with explicit items on human factors and safety (Vasey et al., 2022); CONSORT-AI for randomised trials (Liu et al., 2020); TRIPOD+AI for prediction models (Collins et al., 2024); and STARD-AI for diagnostic accuracy studies (Sounderajah et al., 2025). Metrics Reloaded recommends combining overlap and boundary metrics for segmentation and calibration metrics when probabilities are used (Maier-Hein et al., 2024). A scoping review of 86 randomised trials of AI found

mostly single-centre studies with diagnostic-yield endpoints and poor demographic reporting (Han et al., 2024).

2.6.2 Does the team beat its parts?

The evidence is mixed and instructive. On average, human-AI combinations underperformed the better of human or AI alone in decision tasks (Malone et al., 2024). The effect of AI assistance varied greatly between radiologists, could not be predicted from experience, and incorrect AI predictions harmed performance (Yu et al., 2024); clinicians were susceptible to inaccurate advice whatever its claimed source (Gaube et al., 2021). Explanations increased reliance on wrong advice (Bansal et al., 2021) and had variable effects across clinicians (Nicolson et al., 2025). Positive results come where the human can weigh the AI's reliability case by case: endoscopists followed AI more when it was right than when it was wrong, and the team beat both (Reverberi et al., 2022). Cognitive forcing functions reduced over-reliance at a cost in perceived usability (Bucinca et al., 2021). Most relevant to this study, selective prediction reduced the overall harm of automation bias but changed the error pattern: when told the AI was abstaining, clinicians missed more diagnoses and treatments (Jabbour et al., 2025). Automation bias also appears under time pressure in computational pathology (Rosbach et al., 2024) and in neuroradiology reading (Kim et al., 2025).

2.6.3 Workload, usability and trust

A meta-analysis found that concurrent AI assistance reduced reading time by 27%, and AI pre-screening reduced reading volume by 62%, with sensitivity maintained or improved (Chen et al., 2024a). In mammography, automatic identification of normal examinations cut workload by 17% while excluding 1% of true positives (Rodriguez-Ruiz et al., 2019), and AI-integrated screening pathways improved single-reader performance (Frazer et al., 2024). In a low- and middle-income radiotherapy setting, AI auto-segmentation reduced contouring from about an hour to ten minutes of human review (Kibudde et al., 2024). Usability and trust are measured with established instruments—the System Usability Scale, NASA Task Load Index and Likert trust scales (Brooke, 1996; Hart & Staveland, 1988)—and trust is shaped by clinical experience, perceived risk and how uncertainty is presented (Cao et al., 2024).

2.6.4 Surgical planning and neurosurgery

AI in neurosurgery is reviewed broadly (Awuah et al., 2024; Baker et al., 2024), and machine-learning-based trajectory planning to the tumour has been demonstrated (Dundar et al., 2022). Navigated tractography is accurate for motor and language pathways in glioma surgery (Ille et al., 2021), and automated post-operative segmentation can standardise extent-of-resection assessment (Luque et al., 2024; Zanier et al., 2023). These works evaluate geometric accuracy rather than surgeon acceptability, and none presents the uncertainty of the underlying anatomy alongside a recommended route.

2.6.5 Gap for O4

No reporting standard defines how to measure the quality of an escalation decision itself—whether the system deferred on the cases that needed a clinician and acted on the ones that did not—and evidence on how clinicians respond to a deferring system in brain-tumour care is absent. Evaluations in low-resource settings are rare. A multi-reader, multi-case evaluation that measures escalation appropriateness, missed escalations, workload, usability and trust in situ is the contribution of O4 and RQ4.

2.7 Synthesis

Four things are established. The perception tasks the framework requires are technically solved to a clinically useful standard on curated data, for MRI (Isensee et al., 2021; Pemberton et al., 2023) and for digitised slides (Lu et al., 2021; Wang et al., 2023). Tools for quantifying and calibrating uncertainty are mature (Guo et al., 2017; Lakshminarayanan et al., 2017; Angelopoulos & Bates, 2021), with formal mechanisms for acting on it (Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020). Federated learning is a proven route to multi-institutional training without data sharing (Pati et al., 2022; Lee et al., 2024). And both ethical guidance and regulation require demonstrable human oversight of high-risk clinical AI (World Health Organization, 2021; European Union, 2024).

Four things remain open, and together they define this research (Table 2.3). The primitives for selective autonomy have not been assembled into a calibrated escalation policy for a dependent chain of clinical sub-decisions with asymmetric costs and finite expert capacity. Agentic architectures do not treat uncertainty as part of the contract between agents, so hand-offs are unqualified and accountability for composite decisions is undefined. Multi-source diagnosis is demonstrated on complete, high-quality cohorts rather than under the

incompleteness of real practice. And data-efficient, federated and calibrated learning have been studied separately, leaving unexamined how federated updates and domain adaptation change calibration and thereby silently change the safety behaviour of an escalation policy.

Table 2.3: Summary of open problems and how this research addresses them.

Open problem	Key evidence	Addressed by
Escalation policy for a dependent chain with asymmetric costs and finite clinician time	(Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020; Alves et al., 2024; Dvijotham et al., 2023)	O2 / RQ2 (WP3)
Uncertainty as part of the contract between agents; auditable hand-offs	(Cemri et al., 2025; Zhu et al., 2025; Li et al., 2026; Liu et al., 2026)	O1 / RQ1 (WP3, WP1)
Fusion and perception robust to missing, poor or local data	(Chen et al., 2022; Saueressig et al., 2026; Adewole et al., 2026; Boyd et al., 2024)	O3 / RQ3 (WP2, WP4)
Effect of federation and adaptation on calibration and deferral	(Manthe et al., 2024; Ovadia et al., 2019; Koutsoubis et al., 2024)	O3 / RQ3 (WP5)
Measuring escalation quality, workload and trust in situ	(Vasey et al., 2022; Malone et al., 2024; Jabbour et al., 2025)	O4 / RQ4 (WP6, WP7)

References

Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., & Zhang, L. (2016). Deep learning with differential privacy. ACM SIGSAC Conference on Computer and Communications Security. https://doi.org/10.1145/2976749.2978318

Adewole M, Rudie JD, Gbadamosi A, et al. (2026). Brain tumor segmentation in Sub-Saharan Africa patient population: The BraTS-Africa challenge. Neuro-Oncology Advances. https://doi.org/10.1093/noajnl/vdag082

Adnan M, Kalra S, Cresswell JC, Taylor GW, Tizhoosh HR (2022). Federated learning and differential privacy for medical image analysis. Scientific reports. https://doi.org/10.1038/s41598-022-05539-7

Alshamlan H (2026). Uncertainty-Aware Selective Prediction in Brain-Tumour MRI Classification: Selective Risk Reduction and Component-Wise Reliability Findings. Bioengineering. https://doi.org/10.3390/bioengineering13101184

Alves JV, Leitao D, Jesus S, Sampaio MOP, Liebana J, Saleiro P, et al. (2024). Cost-Sensitive Learning to Defer to Multiple Experts with Workload Constraints. Transactions on Machine Learning Research (arXiv preprint). https://doi.org/10.48550/arxiv.2403.06906

Angelopoulos AN, Bates S (2021). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv 2107.07511 (tutorial). https://arxiv.org/abs/2107.07511

Angelopoulos AN, Bates S, Fisch A, Lei L, Schuster T (2024a). Conformal Risk Control. ICLR 2024. https://doi.org/10.48550/arXiv.2208.02814

Angelopoulos AN, Pomerantz S, Do S, Bates S, et al. (2024b). Conformal Triage for Medical Imaging AI Deployment. medRxiv. https://doi.org/10.1101/2024.02.09.24302543

Ankomah CT, Ayivor LE, Nyame I, et al. (2025). How We Won BraTS-SSA 2025: Brain Tumor Segmentation in the Sub-Saharan African Population Using Segmentation-Aware Data Augmentation and Model Ensembling. arXiv (MICCAI BrainLes 2025, LNCS). https://arxiv.org/abs/2510.03568

Awuah WA, Adebusoye FT, Wellington J, David L, Salam A, Weng Yee AL, et al. (2024). Recent Outcomes and Challenges of Artificial Intelligence, Machine Learning, and Deep Learning in Neurosurgery. World neurosurgery: X. https://doi.org/10.1016/j.wnsx.2024.100301

Baid U, Ghodasara S, Mohan S, Bilello M, Calabrese E, Bakas S, et al. (2021). The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification. arXiv 2107.02314. https://arxiv.org/abs/2107.02314

Baker CR, Pease M, Sexton DP, Abumoussa A, Chambless LB (2024). Artificial intelligence innovations in neurosurgical oncology: a narrative review. Journal of neuro-oncology. https://doi.org/10.1007/s11060-024-04757-5

Bangalore Yogananda CG, Truong NCD, Wagner BC, et al. (2025). Bridging the clinical gap: Confidence informed IDH prediction in brain gliomas using MRI and deep learning. Neuro-Oncology Advances. https://doi.org/10.1093/noajnl/vdaf142

Bansal G, Wu T, Zhou J, Fok R, Nushi B, Kamar E, et al. (2021). Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance. ACM CHI 2021. https://arxiv.org/abs/2006.14779

Bedi S, Mlauzi I, Shin D, Koyejo S, Shah NH (2025). The Optimization Paradox in Clinical AI Multi-Agent Systems. arXiv preprint. https://arxiv.org/abs/2506.06574

Bernhardt M, Ribeiro FDS, Glocker B (2022). Failure Detection in Medical Image Classification: A Reality Check and Benchmarking Testbed. Transactions on Machine Learning Research. https://arxiv.org/abs/2205.14094

Boyd A, Ye Z, Prabhu SP, Tjong MC, Zha Y, et al. (2024). Stepwise Transfer Learning for Expert-level Pediatric Brain Tumor MRI Segmentation in a Limited Data Scenario. Radiology: Artificial Intelligence. https://doi.org/10.1148/ryai.230254

Brooke, J. (1996). SUS: a quick and dirty usability scale. In Usability evaluation in industry (pp. 189-194). Taylor and Francis.

Bucinca Z, Malaya MB, Gajos KZ (2021). To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. Proceedings of the ACM on Human-Computer Interaction (CSCW). https://doi.org/10.1145/3449287

Bungert TJ, Kobelke L, Jaeger PF (2023). Understanding Silent Failures in Medical Image Classification. MICCAI 2023. https://doi.org/10.48550/arXiv.2307.14729

Campanella G, Chen S, Singh M, Verma R, Muehlstedt S, Zeng J, et al. (2025). A clinical benchmark of public self-supervised pathology foundation models. Nature communications. https://doi.org/10.1038/s41467-025-58796-1

Cao S, Liu A, Huang C (2024). Designing for Appropriate Reliance: The Roles of AI Uncertainty Presentation, Initial User Decision, and User Demographics in AI-Assisted Decision-Making. arXiv preprint. https://arxiv.org/abs/2401.05612

Cemri M, Pan MZ, Yang S, Agrawal LA, Chopra B, Tiwari R, et al. (2025). Why Do Multi-Agent LLM Systems Fail?. NeurIPS 2025 Datasets and Benchmarks Track (arXiv:2503.13657). https://arxiv.org/abs/2503.13657

Chen M, Wang Y, Wang Q, et al. (2024a). Impact of human and artificial intelligence collaboration on workload reduction in medical image interpretation. npj Digital Medicine. https://doi.org/10.1038/s41746-024-01328-w

Chen RJ, Ding T, Lu MY, Williamson DFK, Jaume G, Mahmood F, et al. (2024b). Towards a general-purpose foundation model for computational pathology (UNI). Nature Medicine. https://arxiv.org/abs/2308.15474

Chen RJ, Lu MY, Wang J, Williamson DFK, Rodig SJ, Lindeman NI, et al. (2022). Pathomic Fusion: An Integrated Framework for Fusing Histopathology and Genomic Features for Cancer Diagnosis and Prognosis. IEEE Transactions on Medical Imaging. https://arxiv.org/abs/1912.08937

Chen X, Yi H, You M, Liu W, Wang L, Li H, et al. (2025). Enhancing diagnostic capability with multi-agents conversational large language models. npj Digital Medicine (2025) 8:159. https://doi.org/10.1038/s41746-025-01550-0

Ciecierski-Holmes T, Singh R, Axt M, et al. (2022). Artificial intelligence for strengthening healthcare systems in low- and middle-income countries: a systematic scoping review. npj Digital Medicine. https://doi.org/10.1038/s41746-022-00700-y

Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Logullo P, et al. (2024). TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. https://doi.org/10.1136/bmj-2023-078378

Dai Q, Wei D, Liu H, et al. (2024). Federated Modality-Specific Encoders and Multimodal Anchors for Personalized Brain Tumor Segmentation. AAAI Conference on Artificial Intelligence. https://doi.org/10.1609/aaai.v38i2.27909

Dolezal JM, Srisuwananukorn A, Karpeyev D, Ramesh S, et al. (2022). Uncertainty-informed deep learning models enable high-confidence predictions for digital histopathology. Nature Communications. https://doi.org/10.1038/s41467-022-34025-x

Dundar TT, Yurtsever I, Pehlivanoglu MK, Yildiz U, Eker A, Demir MA, et al. (2022). Machine Learning-Based Surgical Planning for Neurosurgery: Artificial Intelligent Approaches to the Cranium. Frontiers in surgery. https://doi.org/10.3389/fsurg.2022.863633

Dvijotham K, Winkens J, Barsbey M, Ghaisas S, Stanforth R, Pawlowski N, et al. (2023). Enhancing the reliability and accuracy of AI-enabled diagnosis via complementarity-driven deferral to clinicians. Nature Medicine 29(7):1814-1820. https://doi.org/10.1038/s41591-023-02437-x

Elyassirad D, Gheiji B, Vatanparast M, Ahmadzadeh AM, et al. (2025). CONSeg: Voxelwise Uncertainty Quantification for Glioma Segmentation Using Conformal Prediction. American Journal of Neuroradiology. https://doi.org/10.3174/ajnr.A8914

European Union. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Article 14: Human oversight. Official Journal of the European Union. https://eur-lex.europa.eu/eli/reg/2024/1689/oj

Fallahpour A, Ma J, Munim A, Lyu H, Wang B (2025). MedRAX: Medical Reasoning Agent for Chest X-ray. Proceedings of the 42nd International Conference on Machine Learning (ICML 2025), PMLR 267 (arXiv:2502.02673v2). https://arxiv.org/abs/2502.02673

Ferber D, El Nahhas OSM, Wolflein G, Wiest IC, Clusmann J, Kather JN, et al. (2025). Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology. Nature Cancer 2025 (preprint arXiv 2404.04667). https://arxiv.org/abs/2404.04667

Ferber D, Hilgers L, Höper C, Kinny-Köster B, Eckardt JN, Egger-Heidrich K, et al. (2026). Towards autonomous medical artificial intelligence agents. Nature, 655:1282. https://doi.org/10.1038/s41586-026-10675-5

Fisch A, Jaakkola T, Barzilay R (2022). Calibrated Selective Classification. Transactions on Machine Learning Research. https://doi.org/10.48550/arXiv.2208.12084

Frazer HML, Peña-Solorzano CA, Kwok CF, Elliott MS, Chen Y, Wang C, et al. (2024). Comparison of AI-integrated pathways with human-AI interaction in population mammographic screening for breast cancer. Nature communications. https://doi.org/10.1038/s41467-024-51725-8

Gal Y, Ghahramani Z (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. ICML 2016. https://arxiv.org/abs/1506.02142

Gaube S, Suresh H, Raue M, et al. (2021). Do as AI say: susceptibility in deployment of clinical decision-aids. npj Digital Medicine. https://doi.org/10.1038/s41746-021-00385-9

Geifman Y, El-Yaniv R (2017). Selective Classification for Deep Neural Networks. NeurIPS 2017. https://arxiv.org/abs/1705.08500

Grande HZ, Pitol JG, Schuck LB, Serenato RV, Laroca R, Hochuli AG (2026). On the Role of MRI Sequences in Cross-Dataset Generalization for Brain Tumor Segmentation. arXiv preprint. https://arxiv.org/abs/2608.29944

Guo C, Pleiss G, Sun Y, Weinberger KQ (2017). On Calibration of Modern Neural Networks. ICML 2017. https://arxiv.org/abs/1706.04599

Hager P, Jungmann F, al. RH, et al. (2024). Evaluation and mitigation of the limitations of large language models in clinical decision-making. Nature Medicine. https://doi.org/10.1038/s41591-024-03097-1

Han R, Acosta JN, Shakeri Z, et al. (2024). Randomised controlled trials evaluating artificial intelligence in clinical practice: a scoping review. The Lancet Digital Health. https://doi.org/10.1016/S2589-7500(24)00047-5

Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (Task Load Index): results of empirical and theoretical research. Advances in Psychology, 52, 139-183. https://doi.org/10.1016/S0166-4115(08)62386-9

Hatamizadeh A, Nath V, Tang Y, Yang D, Roth HR, Xu D (2022). Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images. MICCAI BrainLes Workshop 2021. https://arxiv.org/abs/2201.01266

Hauke F, Wienholt P, Kuhl C, Ferber D, Kather JN, Nebelung S, et al. (2026). Bayesian uncertainty estimation improves clinical decision making in medical AI agents. arXiv preprint. https://arxiv.org/abs/2607.20582

Hollon T, Jiang C, Chowdury A, Nasir-Moin M, et al. (2023). Artificial-intelligence-based molecular classification of diffuse gliomas using rapid, label-free optical imaging. Nature Medicine. https://doi.org/10.1038/s41591-023-02252-4

Hong S, Zhuge M, Chen J, Zheng X, Cheng Y, Zhang C, et al. (2023). MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework. International Conference on Learning Representations (ICLR 2024). https://arxiv.org/abs/2308.00352

Hüllermeier E, Waegeman W (2021). Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. Machine Learning. https://doi.org/10.1007/s10994-021-05946-3

Ille S, Schwendner M, Zhang W, Schroeder A, Meyer B, Krieg SM (2021). Tractography for Subcortical Resection of Gliomas Is Highly Accurate for Motor and Language Function: ioMRI-Based Elastic Fusion Disproves the Severity of Brain Shift. Cancers. https://doi.org/10.3390/cancers13081787

Ilse M, Tomczak JM, Welling M (2018). Attention-based Deep Multiple Instance Learning. ICML 2018. https://arxiv.org/abs/1802.04712

International Agency for Research on Cancer. (2024). GLOBOCAN 2024: Sri Lanka fact sheet. Global Cancer Observatory, IARC/WHO. https://gco.iarc.who.int/media/globocan/factsheets/populations/144-sri-lanka-fact-sheet.pdf

Isensee F, Jaeger PF, Kohl SAA, Petersen J, Maier-Hein KH (2021). nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nature Methods. https://arxiv.org/abs/1809.10486

Isensee F, Wald T, Ulrich C, et al. (2024). nnU-Net Revisited: A Call for Rigorous Validation in 3D Medical Image Segmentation. MICCAI 2024 (LNCS). https://doi.org/10.1007/978-3-031-72114-4_47

Jabal MS, Zhang J, LaBella D, Houk JL, Zhang D, Rudie JD, et al. (2026). Agentic Automation of BT-RADS Scoring: End-to-End Multi-Agent System for Standardized Brain Tumor Follow-up Assessment. arXiv preprint. https://arxiv.org/abs/2603.21494

Jabbour S, Fouhey D, Banovic N, Shepard SD, Kazerooni E, Sjoding MW, et al. (2025). Selective Prediction Reduces the Negative Effects of Automation Bias Overall but Increases False Negatives. arXiv preprint. https://arxiv.org/abs/2508.07617

Jaeger PF, Luth CT, Klein L, Bungert TJ (2023). A Call to Reflect on Evaluation Practices for Failure Detection in Image Classification. ICLR 2023. https://arxiv.org/abs/2211.15259

Jiang M, Wang Z, Dou Q (2022). HarmoFL: Harmonizing Local and Global Drifts in Federated Learning on Heterogeneous Medical Images. AAAI Conference on Artificial Intelligence. https://doi.org/10.1609/aaai.v36i1.19993

Joshi S, Parbhoo S, Doshi-Velez F (2022). Learning-to-defer for sequential medical decision-making under uncertainty. Transactions on Machine Learning Research. https://arxiv.org/abs/2109.06312

Karargyris A, Umeton R, Sheller MJ, Aristizabal A, et al. (2023). Federated benchmarking of medical artificial intelligence with MedPerf. Nature Machine Intelligence. https://doi.org/10.1038/s42256-023-00652-2

Karimireddy, S. P., Kale, S., Mohri, M., Reddi, S., Stich, S., & Suresh, A. T. (2020). SCAFFOLD: stochastic controlled averaging for federated learning. Proceedings of the 37th International Conference on Machine Learning, PMLR 119, 5132-5143.

Kibudde S, Kavuma A, Hao Y, et al. (2024). Impact of Artificial Intelligence-Based Autosegmentation of Organs at Risk in Low- and Middle-Income Countries. Advances in Radiation Oncology. https://doi.org/10.1016/j.adro.2024.101638

Kim SH, Schramm S, Riedel EO, Schmitzer L, Rosenkranz E, Kertels O, et al. (2025). Automation bias in AI-assisted detection of cerebral aneurysms on time-of-flight MR angiography. La Radiologia medica. https://doi.org/10.1007/s11547-025-01964-6

Kim Y, Park C, Jeong H, Chan YS, Xu X, McDuff D, et al. (2024). MDAgents: An Adaptive Collaboration of LLMs for Medical Decision-Making. NeurIPS 2024. https://arxiv.org/abs/2404.15155

Kirchhof M, Kasneci G, Kasneci E (2025). Position: Uncertainty Quantification Needs Reassessment for Large-language Model Agents. Proceedings of the 42nd International Conference on Machine Learning (ICML 2025), Position Paper Track, PMLR 267, Vancouver, Canada. https://arxiv.org/abs/2505.22655

Klang E, Omar M, Raut G, Agbareia R, Timsina P, Freeman R, et al. (2025). Orchestrated multi agents sustain accuracy under clinical-scale workloads compared to a single agent. medRxiv : the preprint server for health sciences. https://doi.org/10.1101/2025.08.22.25334049

Koutsoubis N, Waqas A, Yilmaz Y, Ramachandran RP, et al. (2025). Privacy-preserving Federated Learning and Uncertainty Quantification in Medical Imaging. Radiology: Artificial Intelligence. https://doi.org/10.1148/ryai.240637

Koutsoubis N, Waqas A, Yilmaz Y, Ramachandran RP, Schabath M, Rasool G (2024). Future-Proofing Medical Imaging with Privacy-Preserving Federated Learning and Uncertainty Quantification: A Review. arXiv preprint. https://arxiv.org/abs/2409.16340

Kwon H, Kim DJ (2026). Conformal selective prediction with cost aware deferral for safe clinical triage under distribution shift. Scientific Reports. https://doi.org/10.1038/s41598-026-40637-w

Kömen J, Marienwald H, Dippel J, Hense J (2024). Do Histopathological Foundation Models Eliminate Batch Effects? A Comparative Study. arXiv preprint. https://arxiv.org/abs/2411.05489

Lakshminarayanan B, Pritzel A, Blundell C (2017). Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. NeurIPS 2017. https://arxiv.org/abs/1612.01474

Lambert B, Forbes F, Doyle S, Dehaene H, Dojat M (2024). Trustworthy clinical AI solutions: A unified review of uncertainty quantification in Deep Learning models for medical image analysis. Artificial Intelligence in Medicine. https://doi.org/10.1016/j.artmed.2024.102830

Lee EH, Han M, Wright J, Kuwabara M, Mevorach J, et al. (2024). An international study presenting a federated learning AI platform for pediatric brain tumors. Nature Communications. https://doi.org/10.1038/s41467-024-51172-5

Lekadir K, Frangi AF, Porras AR, et al. (2025). FUTURE-AI: international consensus guideline for trustworthy and deployable artificial intelligence in healthcare. BMJ. https://doi.org/10.1136/bmj-2024-081554

Li T, Sahu AK, Zaheer M, Sanjabi M, Talwalkar A, Smith V (2020). Federated Optimization in Heterogeneous Networks (FedProx). MLSys 2020. https://arxiv.org/abs/1812.06127

Li X, Jiang M, Xu G, Xia J, Chiu C, Jia M, et al. (2026). CRC-Router: Risk-Constrained Routing for Medical Agentic AI Systems. arXiv preprint. https://arxiv.org/abs/2609.30714

Li X, Jiang M, Zhang X, Kamp M, Dou Q (2021). FedBN: Federated Learning on Non-IID Features via Local Batch Normalization. ICLR 2021. https://arxiv.org/abs/2102.07623

Linardos A, Pati S, Baid U, et al. (2025). The MICCAI Federated Tumor Segmentation (FeTS) Challenge 2024: Efficient and Robust Aggregation Methods. Machine Learning for Biomedical Imaging (MELBA). https://doi.org/10.59275/j.melba.2025-5242

Liu F, Niu Y, Zhang Q, Wang K, Dong Z, Wong IN, et al. (2025). A foundational architecture for AI agents in healthcare. Cell Reports Medicine. https://doi.org/10.1016/j.xcrm.2025.102374

Liu H, Wei D, Lu D, et al. (2023). M3AE: Multimodal Representation Learning for Brain Tumor Segmentation with Missing Modalities. AAAI Conference on Artificial Intelligence. https://doi.org/10.1609/aaai.v37i2.25253

Liu J, Gallego B, Barbieri S (2022). Incorporating uncertainty in learning to defer algorithms for safe computer-aided diagnosis. Scientific Reports. https://doi.org/10.1038/s41598-022-05725-7

Liu X, Cruz Rivera S, Moher D, Calvert MJ, SPIRIT-AI DA, Group CW (2020). Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension. Nature Medicine. https://doi.org/10.1038/s41591-020-1034-x

Liu Y, Zhang Z, Liu R, Xitang M, Shin S, Lee H (2026). TumorBoard: Evidence-Grounded Multi-Agent Decision Support for Longitudinal Neuro-Oncology. arXiv preprint. https://arxiv.org/abs/2608.03190

Loftus TJ, Shickel B, Ruppert MM, Balch JA, et al. (2022). Uncertainty-aware deep learning in healthcare: A scoping review. PLOS Digital Health. https://doi.org/10.1371/journal.pdig.0000085

Lou J, Yang Y, Yu Z, Fu Z, Han W, Huang Q, et al. (2026). CXRAgent: Director-Orchestrated Multi-Stage Reasoning for Chest X-Ray Interpretation. IEEE Transactions on Medical Imaging (accepted version read: arXiv:2510.21324v2). https://doi.org/10.1109/tmi.2026.3713585

Louis, D. N., Perry, A., Wesseling, P., et al. (2021). The 2021 WHO Classification of Tumors of the Central Nervous System: a summary. Neuro-Oncology, 23(8), 1231-1251. https://doi.org/10.1093/neuonc/noab106

Lu C, Yu Y, Karimireddy SP, Jordan MI, Raskar R (2023). Federated Conformal Predictors for Distributed Uncertainty Quantification. arXiv preprint. https://arxiv.org/abs/2305.17564

Lu MY, Williamson DFK, Chen TY, Chen RJ, Barbieri M, Mahmood F (2021). Data-efficient and weakly supervised computational pathology on whole-slide images (CLAM). Nature Biomedical Engineering. https://arxiv.org/abs/2004.09666

Luque L, Skogen K, MacIntosh BJ, Emblem KE, Larsson C, Bouget D, et al. (2024). Standardized evaluation of the extent of resection in glioblastoma with automated early post-operative segmentation. Frontiers in radiology. https://doi.org/10.3389/fradi.2024.1357341

Ma J, He Y, Li F, Han L, You C, Wang B (2024). Segment Anything in Medical Images (MedSAM). Nature Communications. https://arxiv.org/abs/2304.12306

Ma J, Yang Z, Kim S, et al. (2025). MedSAM2: Segment Anything in 3D Medical Images and Videos. arXiv. https://arxiv.org/abs/2504.03600

Madras D, Pitassi T, Zemel R (2018). Predict Responsibly: Improving Fairness and Accuracy by Learning to Defer. NeurIPS 2018. https://arxiv.org/abs/1711.06664

Maier-Hein L, Reinke A, Godau P, et al. (2024). Metrics reloaded: recommendations for image analysis validation. Nature Methods. https://doi.org/10.1038/s41592-023-02151-z

Malone TW, Vaccaro MA, al. AC, et al. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. Nature Human Behaviour. https://doi.org/10.1038/s41562-024-02024-1

Manthe M, Duffner S, Lartizien C (2024). Federated brain tumor segmentation: An extensive benchmark. Medical Image Analysis. https://doi.org/10.1016/j.media.2024.103270

McMahan HB, Moore E, Ramage D, Hampson S, Aguera y Arcas B (2017). Communication-Efficient Learning of Deep Networks from Decentralized Data (FedAvg). AISTATS 2017. https://arxiv.org/abs/1602.05629

Mehrtash A, Wells WM, Tempany CM, Abolmaesumi P, Kapur T (2020). Confidence Calibration and Predictive Uncertainty Estimation for Deep Medical Image Segmentation. IEEE Transactions on Medical Imaging. https://arxiv.org/abs/1911.13273

Mehrtens HA, Bucher T, Brinker TJ (2023a). Pitfalls of Conformal Predictions for Medical Image Classification. UNSURE Workshop at MICCAI 2023 (Lecture Notes in Computer Science). https://doi.org/10.1007/978-3-031-44336-7_20

Mehrtens HA, Kurz A, Bucher TC, Brinker TJ (2023b). Benchmarking common uncertainty estimation methods with histopathological images under domain shift and label noise. Medical Image Analysis. https://doi.org/10.1016/j.media.2023.102914

Mehta R, Filos A, Baid U, Sako C, McKinley R, Rebsamen M, et al. (2022). QU-BraTS: MICCAI BraTS 2020 Challenge on Quantifying Uncertainty in Brain Tumor Segmentation. Machine Learning for Biomedical Imaging (MELBA). https://arxiv.org/abs/2112.10074

Mitchell M, Wu S, Zaldivar A, Barnes P, Vasserman L, Hutchinson B, et al. (2019). Model Cards for Model Reporting. ACM FAT* 2019. https://arxiv.org/abs/1810.03993

Molinaro, A. M., Hervey-Jumper, S., Morshed, R. A., et al. (2020). Association of maximal extent of resection of contrast-enhanced and non-contrast-enhanced tumor with survival within molecular subgroups of patients with newly diagnosed glioblastoma. JAMA Oncology, 6(4), 495-503. https://doi.org/10.1001/jamaoncol.2019.6143

Mozannar H, Sontag D (2020). Consistent Estimators for Learning to Defer to an Expert. ICML 2020. https://arxiv.org/abs/2006.01862

National Cancer Control Programme. (2024). National cancer incidence and mortality data, Sri Lanka. Ministry of Health, Sri Lanka. https://www.nccp.health.gov.lk/

Nicolson A, Bradburn E, Gal Y, et al. (2025). The human factor in explainable artificial intelligence: clinician variability in trust, reliance, and performance. npj Digital Medicine. https://doi.org/10.1038/s41746-025-02023-0

Okati N, De A, Gomez-Rodriguez M (2021). Differentiable Learning Under Triage. NeurIPS 2021. https://arxiv.org/abs/2103.08902

Olsson H, Kartasalo K, Mulliqi N, Capuccini M, et al. (2022). Estimating diagnostic uncertainty in artificial intelligence assisted pathology using conformal prediction. Nature Communications. https://doi.org/10.1038/s41467-022-34945-8

Ovadia Y, Fertig E, Ren J, Nado Z, Sculley D, Nowozin S, et al. (2019). Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift. NeurIPS 2019. https://arxiv.org/abs/1906.02530

Pati S, Baid U, Edwards B, Sheller M, Wang SH, Bakas S, et al. (2022). Federated learning enables big data for rare cancer boundary detection. Nature Communications. https://arxiv.org/abs/2204.10836

Pemberton HG, Wu J, Kommers I, Muller DMJ, Hu Y, et al. (2023). Multi-class glioma segmentation on real-world data with missing MRI sequences: comparison of three deep learning algorithms. Scientific Reports. https://doi.org/10.1038/s41598-023-44794-0

Price, M., Ostrom, Q. T., et al. (2024). CBTRUS statistical report: primary brain and other central nervous system tumors diagnosed in the United States in 2017-2021. Neuro-Oncology, 26(Suppl. 6), vi1-vi85. https://doi.org/10.1093/neuonc/noae145

Raghu M, Blumer K, Corrado G, Kleinberg J, Obermeyer Z, Mullainathan S (2019). The Algorithmic Automation Problem: Prediction, Triage, and Human Effort. arXiv 1903.12220. https://arxiv.org/abs/1903.12220

Raji ID, Smart A, White RN, Mitchell M, Gebru T, Hutchinson B, et al. (2020). Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing. ACM FAT* 2020. https://arxiv.org/abs/2001.00973

Reverberi C, Rigon T, Solari A, et al. (2022). Experimental evidence of effective human-AI collaboration in medical decision-making. Scientific Reports. https://doi.org/10.1038/s41598-022-18751-2

Rivera JEH, Low DK, Xiong X, Ruzevick JJ, Child DD, Yim W, et al. (2026). CoRe-BT: A Multimodal Radiology-Pathology-Text Benchmark for Robust Brain Tumor Typing. arXiv preprint. https://arxiv.org/abs/2603.03618

Rodriguez-Ruiz A, Lång K, Gubern-Merida A, Teuwen J, Broeders M, Gennaro G, et al. (2019). Can we reduce the workload of mammographic screening by automatic identification of normal exams with artificial intelligence? A feasibility study. European radiology. https://doi.org/10.1007/s00330-019-06186-9

Rosbach E, Ganz J, Ammeling J, Riener A, Aubreville M (2024). Automation Bias in AI-Assisted Medical Decision-Making under Time Pressure in Computational Pathology. arXiv preprint. https://arxiv.org/abs/2411.00998

Roy S, Koehler G, Ulrich C, et al. (2023). MedNeXt: Transformer-Driven Scaling of ConvNets for Medical Image Segmentation. MICCAI 2023 (LNCS). https://doi.org/10.1007/978-3-031-43901-8_39

Sanai, N., & Berger, M. S. (2008). Glioma extent of resection and its impact on patient outcome. Neurosurgery, 62(4), 753-766. https://doi.org/10.1227/01.neu.0000318159.21731.cf

Saueressig C, Scholz D, Raffler P, Delbridge C, et al. (2026). Multimodal fusion of pathology and radiology foundation models for WHO 2021 glioma subtyping. npj Precision Oncology. https://doi.org/10.1038/s41698-026-01366-5

Sharma K, Sivadas H, Edwards C, Gunja N, Reddy S (2026). ED-triage-agent: a multi-agent framework for human-in-the-loop emergency triage. International Journal of Medical Informatics. https://doi.org/10.1016/j.ijmedinf.2026.106652

Sheller MJ, Edwards B, Reina GA, Martin J, Pati S, Kotrotsou A, et al. (2020). Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. Scientific Reports. https://doi.org/10.1038/s41598-020-69250-1

Sounderajah V, Guni A, al. XL, et al. (2025). The STARD-AI reporting guideline for diagnostic accuracy studies using artificial intelligence. Nature Medicine. https://doi.org/10.1038/s41591-025-03953-8

Tang X, Zou A, Zhang Z, Li Z, Zhao Y, Zhang X, et al. (2024). MedAgents: Large Language Models as Collaborators for Zero-shot Medical Reasoning. Findings of ACL 2024. https://arxiv.org/abs/2311.10537

Tang Y, Yang D, Li W, Roth HR, Landman B, Xu D, et al. (2022). Self-Supervised Pre-Training of Swin Transformers for 3D Medical Image Analysis. CVPR 2022. https://arxiv.org/abs/2111.14791

Vasey B, Nagendran M, Campbell B, Clifton DA, Collins GS, McCulloch P, et al. (2022). Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. Nature Medicine. https://doi.org/10.1038/s41591-022-01772-9

Verma R, Barrejon D, Nalisnick E (2023). Learning to Defer to Multiple Experts: Consistent Surrogate Losses, Confidence Calibration, and Conformal Ensembles. AISTATS 2023. https://arxiv.org/abs/2210.16955

Verma R, Nalisnick E (2022). Calibrated Learning to Defer with One-vs-All Classifiers. ICML 2022. https://arxiv.org/abs/2202.03673

Vermeulen C, Pages-Gallego M, Kester L, Kranendonk MEG, et al. (2023). Ultra-fast deep-learned CNS tumour classification during surgery. Nature. https://doi.org/10.1038/s41586-023-06615-2

Wang J, Jin Y, Wang L (2022). Personalizing Federated Medical Image Segmentation via Local Calibration. European Conference on Computer Vision (ECCV). https://doi.org/10.1007/978-3-031-19803-8_27

Wang W, Chen C, Ding M, et al. (2021). TransBTS: Multimodal Brain Tumor Segmentation Using Transformer. MICCAI 2021 (LNCS). https://doi.org/10.1007/978-3-030-87193-2_11

Wang W, Zhao Y, Teng L, et al. (2023). Neuropathologist-level integrated classification of adult-type diffuse gliomas using deep learning from whole-slide pathological images. Nature Communications. https://doi.org/10.1038/s41467-023-41195-9

World Health Organization. (2021). Ethics and governance of artificial intelligence for health: WHO guidance. WHO. https://www.who.int/publications/i/item/9789240029200

Wu Q, Bansal G, Zhang J, Wu Y, Li B, Zhu E, et al. (2023). AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. arXiv 2308.08155 (COLM 2024). https://arxiv.org/abs/2308.08155

Wu X, Zhang H, Garduno-Rapp NE, Rousseau JF, Thakkallapally M, Ji Y, et al. (2026). Orchestrator multi-agent clinical decision support system for secondary headache diagnosis in primary care. Journal of the American Medical Informatics Association (JAMIA), ocag111 (preprint arXiv:2512.04207). https://doi.org/10.1093/jamia/ocag111

Yao S, Zhao J, Yu D, Du N, Shafran I, Narasimhan K, et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023. https://arxiv.org/abs/2210.03629

Yevudza WE, Buckman V, Darko K, et al. (2024). Neuro-oncology access in Sub-Saharan Africa: A literature review of challenges and opportunities. Neuro-Oncology Advances. https://doi.org/10.1093/noajnl/vdae057

Yu F, Moehring A, Banerjee O, et al. (2024). Heterogeneity and predictors of the effects of AI assistance on radiologists. Nature Medicine. https://doi.org/10.1038/s41591-024-02850-w

Yu Y, Bates S, Ma Y, Jordan MI (2022). Robust Calibration with Multi-domain Temperature Scaling. arXiv preprint. https://arxiv.org/abs/2206.02757

Zanier O, Da Mutten R, Vieli M, Regli L, Serra C, Staartjes VE (2023). DeepEOR: automated perioperative volumetric assessment of variable grade gliomas using deep learning. Acta neurochirurgica. https://doi.org/10.1007/s00701-022-05446-w

Zenk M, Baid U, Pati S, et al. (2025). Towards fair decentralized benchmarking of healthcare AI algorithms with the Federated Tumor Segmentation (FeTS) challenge. Nature Communications. https://doi.org/10.1038/s41467-025-60466-1

Zhang Y, He N, Yang J, et al. (2022). mmFormer: Multimodal Medical Transformer for Incomplete Multimodal Learning of Brain Tumor Segmentation. MICCAI 2022 (LNCS). https://doi.org/10.1007/978-3-031-16443-9_11

Zhao J, Lin SY, Attias R, et al. (2025). Uncertainty-aware ensemble of foundation models differentiates glioblastoma from its mimics. Nature Communications. https://doi.org/10.1038/s41467-025-64249-6

Zhu Y, Gu L, Wang Z, Sang H, Sui D, Tang W, et al. (2025). Auditing medical multi-agent AI reveals risks of false consensus. arXiv preprint. https://arxiv.org/abs/2510.10185